

HOROGEM

Vedic Astrology CRM & Recommendation Suite

TECHNICAL PROJECT NOTES & AI USAGE DOCUMENTATION

Generated on: 13 June 2026

System Version: 1.0.0 (Bilingual Mobile-Responsive Core)

HOROGEM: Developer Project Notes

This document contains key engineering choices, system architecture notes, configuration layouts, and updates completed during the development of **HOROGEM** (Vedic Astrology CRM & Gemstone Recommendation Platform).

◆ System Architecture

The application is structured as a client-side Single Page Application (SPA) built with **Vite**, **React 19**, and **TypeScript**.

1. Calculation Engine (`src/engine/`)

Contains pure, deterministic mathematical and rule-based functions representing classical Vedic astrology:

- **astrology.ts**: Computes planet degrees, sign placements, ascendant (Lagna), moon sign (Rashi), Nakshatras, Classical Yogas (e.g., Budhaditya, Gajakesari), Doshas (Manglik, Sade Sati), and generates 100-year Vimshottari Mahadasha/Antardasha timelines.
- **gemstones.ts**: Implements the **4-step suitability algorithm** to identify Life (Lagnesh), Lucky (Bhagyesh), and Work (Karmesh) stones, and filter out functional malefics or planets placed in Trika houses (6, 8, 12).
- **remedies.ts**: Maps planetary placements to Lal Kitab remedies, donation programs, Rudraksha guidelines, and color therapies.
- **numerology.ts**: Solves psychic numbers (Mulank) and life path numbers (Bhagyank).

2. UI Components (`src/components/`)

- **BirthChart.tsx**: Renders North Indian Kundali grids (D1, D9, D10, Chandra) using SVG. Text elements are placed using responsive Cartesian coordinates.
- **Chatbot.tsx**: Provides an interactive chat interface. Supports fallback keyword matching or live queries using Gemini client-side calls.
- **AstrologerCRM.tsx**: Includes billing ledger, printable invoice models, calendar booking schedulers, and client records.
- **ConsultationRec.tsx**: Captures audio using browser microphone API, renders real-time frequency waveforms on `<canvas>`, and simulates AI-driven session summaries.

◆ Key Engineering Decisions & Updates

1. API Configuration & Key Storage

- **Removed Admin Panel**: The admin panel was removed to simplify navigation and secure configuration access.
- **Global Settings Modal**: The Gemini API key setup is relocated to a global settings popup modal in the header, saved directly in `localStorage` under the key `horogem_gemini_api_key`.
- **Zero-Setup Compatibility**: If no local key is entered, the chatbot falls back to reading the environment variable `VITE_GEMINI_API_KEY` (configured at build-time) or falls back to an offline rule-based Vedic parser.

2. Vercel & GitHub Pages Dynamic Paths

To support simultaneous deployments on both Vercel (root `/` path serving) and GitHub Pages (subfolder `/HOROGEM/` path serving) without manual configuration modifications:

- **vite.config.ts** resolves base path dynamically:

```
base: process.env.GITHUB_ACTIONS ? '/HOROGEM/' : '/'
```

3. Love & Marriage Predictive Logic

- An offline love and arrange marriage calculator is integrated directly into the local rule parser in `Chatbot.tsx`.
- Evaluates planet associations between the **5th House Lord** (romance, courtship), **7th House Lord** (partnership, marriage), Rahu (unconventional/love connections), and Venus/Jupiter (romance and

wisdom signifiers) to predict relationship paths.

4. Full Mobile Responsiveness (Pure CSS & React State Hooks)

- **Overlay Drawer Layout:** On mobile screens (width $\leq$ `768px`), the `360px` sidebar drawer behaves as a fixed overlay drawer rather than squeezing layout viewports. It dims the main interface with a backdrop and collapses automatically when the backdrop is clicked.
 - **Auto-Close calculated profiles:** When details are submitted on a phone, the drawer automatically slides shut to show chart SVG graphics instantly.
 - **Preserved Form States:** The form sidebar contents remain mounted in React (visibility controlled via CSS rules) to prevent resetting user fields during drawer transitions.
 - **Split Two-Row Mobile Header:** Resolves cramped layout headers on small screens by splitting actions (top row) and horizontal-scroll tab selectors (bottom row). Uses modern CSS `display: contents` to keep desktop row alignment seamless.
 - **Horizontal scroll tables:** Wrapped data tables (ledger, matchmaking) in overflow scroll wrappers (`.table-container`) to avoid squishing detailed columns.
 - **Centered inputs dropdowns:** Horizontally centers date and clock pickers to avoid screen margin overflows on narrow widths.
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◆ Local Setup & Production Compilation

- **Type checking check:** `npx tsc --noEmit`
- **Build compiler execution:** `npm run build`
- **Preview static outputs:** `npm run preview`

AI Usage Declaration

This repository has been designed, coded, and tested in a collaborative pair-programming session between a human developer and **Antigravity**, an AI assistant for Advanced Agentic Coding developed by the Google DeepMind team.

◆ AI Contributions

1. Architectural Design & Scaffolding

- Structured the workspace as a Vite-React-TypeScript single page application.
- Configured modular separations between the Vedic mathematics library (`src/engine/`) and the view presentation cards (`src/components/`).
- Designed a custom, responsive dark-cosmic styling system in Vanilla CSS using Custom Properties.
- Created dynamic base path resolution config configurations in `vite.config.ts` to allow automatic deployments on both Vercel (`/`) and GitHub Pages (`/HOROGEM/`) pipelines.

2. Algorithmic Implementations

- Implemented deterministic Vedic astrology calculations, Yogas, and Vimshottari dasha cycles in `astrology.ts` .
- Translated the 4-step gemstone suitability rules (Lagna Lord, Yogakaraka, functional benefic checks, and Trika house filters) into TypeScript logic in `gemstones.ts` .
- Built Chaldean-based name and date numerology solvers in `numerology.ts` .
- Programmed a custom love/marriage predictor engine in `Chatbot.tsx` evaluating relationship connections between the 5th House Lord, 7th House Lord, Rahu, Venus, and Jupiter.

3. Audio & Visual Syntheses

- Programmed a canvas frequency wave visualizer and browser mic capture integrations.
- Built a Web Audio API synthesizer that dynamically generates warm 136.1Hz meditative drone frequencies without external audio file dependencies.

- Rendered North Indian Kundali charts and sacred geometric Yantras using pure SVG paths and canvas line strokes.

4. Layout Optimization & Mobile Responsiveness

- Implemented mobile-responsive overlay drawer layouts for the birth profile manager, enabling smooth slide-in gestures and dimming screen backdrop overlays on mobile screen viewports.
- Structured a flexible double-row header on mobile (top row actions + bottom row tabs) that displays as a single row on desktop using CSS `display: contents` grid/flex overrides.
- Wrapped wide tables in responsive scroll wrappers to eliminate horizontal page breaking.
- Centered time and date pickers to ensure dropdown elements fit screen dimensions on narrow widths.
- Hooked state calculations to automatically slide sidebar panels closed upon form submission on mobile.

5. Settings Configuration & Security Refactoring

- Transferred local configurations from the Admin Panel to a global modal Settings popover accessible in the main header, storing credentials directly in client-side `localStorage`.
- Set up fallback support for the environment variable `VITE_GEMINI_API_KEY` for zero-setup assistant calls.

◆ Developer Oversight

All generated code was compiled, bundle-tested, and audited by the human developer. The project compiles with zero typescript compiler errors (`npm run tsc --noEmit`) and bundles successfully for production deployment.